Sri Lanka Institute of Information Technology

IT2080 – **Progress Report**
Year 2, Semester 2 -2025

Group 03

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1. User Stories Completed and Not Yet Completed

High level user story/ function	Completed (Y/N)	Responsible member
ADMIN MANAGEMENT SYSTEM		
Admin registration and login system	Y	IT23751514
User account management dashboard	Y	
Platform analytics and reporting	N	
Payment transaction monitoring	N	
Content moderation system	Y	
COACH MANAGEMENT SYSTEM		
Coach registration and certification	Y	IT23599222
Availability scheduling system	Y	
Session management tools	N	
Student progress tracking	Y	
Performance analytics	Y	
Uniform Order PDF Generator	Y	
OPERATIONS & INVENTORY MANAGEMENT		
Equipment inventory database	Y	IT23628168
Pricing Rule system	N	
Stock level monitoring	Y	
Equipment allocation system	N	
Ground Managers Approval Handling	Y	
SERVICE PROVIDER MANAGEMENT		
Venue registration and verification	Y	IT23759398
Ground availability management	Y	
Booking confirmation system	N	
Payment processing integration	N	
PLAYER MANAGEMENT		
Player registration and profile management	Y	IT23761278
Booking history and preferences	Y	
Progress tracking dashboard	N	
Team formation tools	N	

Drive Link –
https://drive.google.com/drive/folders/1qCsOB1ahC9RgaKd9rV4eKzeruhdoYAjW?usp=drive_link

ER Diagram Drive Link -
https://drive.google.com/file/d/1lOCz4Jg7KimzkIy1iS6VmFhhjEsw1Dwb/view?usp=drive_link

Normalized Schema Drive Link –
https://drive.google.com/file/d/1pD2sty6u4pdwi3c3jKMChOkOuvOl7SCI/view?usp=drive_link

Network Diagram Drive Link –
https://drive.google.com/file/d/1RSRjHOyS1fyQnVsV8vcksNvPPP2Qqbo9/view?usp=drive_link

System Diagram Drive Link –
https://drive.google.com/file/d/1ufngP8NfM-7mnmMgiqpYu_lPf0hIYqoQ/view?usp=drive_link

2. Innovative Parts of The Project

1. Unified Sports Ecosystem

- Integrated Platform: Single solution for bookings, equipment, coaching, and payments
- Real-time Synchronization: Live availability updates across all services
- Multi-role Dashboard: Tailored interfaces for players, coaches, venues, and admins

2. Advanced Technology Integration

- Mobile-First PWA: Offline capability and app-like experience
- Blockchain Verification: Secure certification validation for coaches and venues
- IoT Equipment Tracking: QR code/RFID monitoring for inventory management

3. User-Centric Features

- Gamified Progress Tracking: Badges and rewards system for player motivation
- Smart Notifications: Context-aware alerts and reminders
- Multi-language Support: English, Sinhala, and Tamil interface

3. Commercialization

Concept Development and Research

The idea for LankaArena - Comprehensive Sports Management Platform emerged from the significant inefficiencies in managing sports activities across Sri Lankan universities, schools, and clubs using fragmented or

non-existent digital systems. Currently, sports teams rely on manual processes, phone calls, social media groups, and paper-based systems for venue bookings, equipment management, and match coordination. This leads to operational chaos, double bookings, financial losses, limited access to resources, and poor communication between athletes, coaches, and venue owners.
Our platform, LankaArena, aims to centralize all sports management operations into a unified digital ecosystem, enabling seamless management of venue bookings, equipment rentals, coach hiring, match scheduling, and player development. By integrating all essential sports management functions into one intuitive system, LankaArena will enhance operational efficiency, improve athlete experiences, and streamline sports administration for educational institutions and sports clubs across Sri Lanka.

Software Product Development

- Built using the MERN Stack (MongoDB, Express.js, React.js, Node.js) for scalability and real-time performance, LankaArena features eight core modules:
- Admin Management System
- Player Management System
- Coach Management System
- Operations & Equipment Management
- Service Providers Management
- Booking & Scheduling Management
- Payment & Financial Management
- Analytics & Reporting System

Revenue Model

- Subscription Plans:
- Basic: Free (limited features)
- Premium: LKR 490/month (advanced features)
- Enterprise: Custom pricing (full system)
- Commission-based:
- Venue bookings: 5-7%
- Equipment rentals: 8-10%
- Coach sessions: 12-15%
- Premium Services: AI matchmaking (LKR 199/month), Advanced analytics (LKR 299/month)

Market Strategy

- Phase 1 (Months 1-6): Target universities and schools through campus ambassadors
  Phase 2 (Months 7-12): Expand to sports clubs and federations
  Phase 3 (Year 2): National scale including corporate wellness programs
- Growth Tactics: Digital marketing, partnerships with sports brands, multi-language support (English/Sinhala/Tamil), and freemium model for user acquisition.

Admin Management | IT23751514 | Herath.H.M.P.P

Module: User Registration & Verification

Module Name	User Registration & Verification
Test Case ID	1
Priority Level	High
Description	Verify new user registration and admin verification process

Prerequisites	Admin account must be active, registration page accessible
Environment	Web Browser

Test Case Table

ID	Test Steps	Test Input	Expected Result	Actual Result	Status	Comments
01	Navigate to registration page	-	Registration form loads successfully	Form loaded with all fields displayed correctly	Pass	UI responsive, loading time <2 seconds
02	Enter valid user details	Name, Email, Phone, Password	Form accepts input, shows validation success	All fields accepted, green validation ticks shown	Pass	Password strength meter working properly
03	Submit registration	-	Account created with "Pending Verification" status	"Registration Successful - Awaiting Verification" message displayed	Pass	Email confirmation sent
04	Admin login to verification panel	Admin credentials	Access verification dashboard successfully	Admin dashboard loaded, verification panel accessible	Pass	Dashboard loaded within 3 seconds
05	Verify new user account	Select user, click "Verify"	User status changes to "Active", notification sent	Status changed to "Active", user notified via email	Pass	Verification timestamp recorded correctly

Module: Platform Analytics Dashboard

Module Name	Platform Analytics Dashboard
Test Case ID	2
Priority Level	Medium
Description	Verify analytics data display and report generation
Prerequisites	Admin logged in, system has operational data
Environment	Web Browser

ID	Test Steps	Test Input	Expected Result	Actual Result	Status	Comments
01	Login as Admin	Admin credentials	Redirect to dashboard	Successfully redirected to admin dashboard	Pass	Session established correctly
02	Navigate to Analytics	Click Analytics tab	Analytics dashboard loads with charts	Dashboard loaded with 4 charts: Users, Revenue, Bookings, Equipment	Pass	Charts rendered using Chart.js library
03	Filter by date range	Select: 2025-01-01 to 2025-03-31	Data updates to show selected period	Data filtered correctly, 45 bookings shown for period	Pass	Date picker working accurately

04	Generate revenue report	Click "Generate Report"	PDF report downloads successfully	PDF generated with filename "Revenue_Report_Mar2025.pdf"	Pass	Report included charts and tabular data
05	View user activity metrics	-	Charts display active users, bookings, revenue	Metrics displayed: 240 active users, 89 bookings, LKR 156,800 revenue	Pass	Real-time data synchronization working

Coach Management System | IT23599222 | Wijethunga A.T.U

Module: Coach Profile Management

Module Name	Coach Profile Management
Test Case ID	1
Priority Level	High
Description	Verify coach registration and profile management
Prerequisites	Registration page accessible, coach documents ready
Environment	Web Browser

Test Case Table

ID	Test Steps	Test Input	Expected Result	Actual Result	Status	Comments
01	Navigate to coach registration	-	Coach registration form loads	Form loaded with 8 input sections	Pass	Progress indicator showing 5 steps
02	Enter coach details	Name, specialization, experience	Form validation passes	Green checkmarks for all required fields	Pass	Auto-save feature working every 30 seconds
03	Upload certifications	PDF certificate files	Files upload successfully	File uploaded (2.3 MB), preview available	Pass	File size validation working (max 5MB)
04	Set availability schedule	Mon-Fri, 4 PM-8 PM	Schedule saved correctly	Weekly calendar updated with available slots	Pass	Conflict detection prevented overlapping

05	Submit registration	-	Profile created, pending admin approval	"Application Submitted - Under Review" message shown	Pass	Application ID: COACH_2025_0047 generated

Module: Session Management

Module Name	Session Management
Test Case ID	2
Priority Level	High
Description	Verify coach session scheduling and tracking
Prerequisites	Coach logged in, approved profile
Environment	Web Browser

ID	Test Steps	Test Input	Expected Result	Actual Result	Status	Comments
01	Coach login	Coach credentials	Access coach dashboard	Dashboard loaded with 3 pending sessions	Pass	Welcome back message personalized
02	View session requests	-	Pending session requests displayed	List showed: 2 cricket, 1 football sessions	Pass	Filter by sport type working
03	Accept session request	Select request, click "Accept"	Session confirmed, player notified	Session status changed to "Confirmed"	Pass	Auto-generated calendar invite sent
04	Update session progress	Session notes, player performance	Progress saved successfully	Progress saved with timestamp	Pass	Rich text editor functioning
05	Mark session complete	Click "Complete Session"	Session status updated, invoice generated	Status: Completed, Invoice: INV_CS_2387 generated	Pass	Automatic payment processing initiated

Operations & Equipment Inventory Management | IT23628168 | Alahakoon.A.M.I.B

Module: Equipment Rental Management

Module Name	Equipment Rental Process
Test Case ID	1
Priority Level	High
Description	Verify complete equipment rental workflow

Prerequisites	Equipment items available, user logged in
Environment	Web Browser

Test Case Table

ID	Test Steps	Test Input	Expected Result	Actual Result	Status	Comments
01	User login	Player credentials	Access player dashboard	Player dashboard loaded successfully	Pass	Welcome message displayed correctly
02	Navigate to Equipment section	Click "Rent Equipment"	Equipment catalog displays available items	Catalog showed 23 available items with images	Pass	Filter by sport category working
03	Select equipment item	Cricket bat, 3 days rental	Item added to cart, price calculated	Added to cart: Cricket Bat - LKR 600 × 3 days = LKR 1800	Pass	Rental calculator functioning accurately
04	Proceed to checkout	-	Payment page loads with order summary	Payment page loaded with order summary and total amount	Pass	Security SSL certificate active
05	Complete payment	Card: 4111 1111 1111 1111, Exp: 12/26, CVV: 123	Booking confirmed, equipment reserved	"Payment Successful - Equipment Reserved" confirmation shown	Pass	Payment gateway integration working

Module: Inventory Management

Module Name	Inventory Management
Test Case ID	2
Priority Level	High
Description	Verify inventory tracking and stock updates
Prerequisites	Admin logged in, inventory items exist
Environment	Web Browser

ID	Test Steps	Test Input	Expected Result	Actual Result	Status	Comments
01	Admin login	Admin credentials	Access admin dashboard	Admin dashboard loaded with	Pass	Quick access

				inventory shortcut		menu functional
02	Navigate to Inventory	Click "Equipment Inventory	Inventory list displays all items	List showed 56 items with pagination (10 per page)	Pass	Search functionality working
03	Add new equipment	Item: Basketball, Qty: 15, Price: LKR 400/day	New item added to inventory	"Basketball added successfully" message displayed	Pass	Auto-generated equipment ID: EQ_BB_0042
04	Update stock level	Item: Football, New Qty: 25	Stock quantity updates successfully	Football quantity changed from 18 to 25	Pass	Audit trail recorded the change
05	Set maintenance status	Item: Tennis racket, Status: Under Maintenance	Item marked as unavailable for rental	Status changed, red "Maintenance" badge displayed	Pass	Automatic email alert sent to maintenance team

Service Providers Management I IT23759398 | Karunarathna.N.T.I.N

Module: Venue Management

Module Name	Venue Registration
Test Case ID	1
Priority Level	High
Description	Verify venue owner registration and venue listing
Prerequisites	Service provider registration page accessible
Environment	Web Browser

Test Case Table

ID	Test Steps	Test Input	Expected Result	Actual Result	Status	Comments
01	Navigate to provider registration	-	Registration form loads	Business registration form loaded	Pass	Step-by-step wizard interface
02	Enter venue details	Name, location, facilities	Form validation passes	All details accepted, map location pinned	Pass	Google Maps integration working

03	Set pricing	Hourly rate: LKR 2500	Pricing saved correctly	Pricing matrix saved for different time slots	Pass	Dynamic pricing rules applied
04	Upload venue photos	JPEG images	Photos upload successfully	All 5 photos uploaded, gallery preview available	Pass	Image compression working (reduced to 500KB each)
05	Submit registration	-	Venue listed, pending admin verification	"Registration Submitted - Verification Pending"	Pass	Estimated verification time: 24-48 hours

Module: Booking Management

Module Name	Booking Management
Test Case ID	2
Priority Level	High
Description	Verify venue booking management by service provider
Prerequisites	Venue owner logged in, venue approved
Environment	Web Browser

ID	Test Steps	Test Input	Expected Result	Actual Result	Status	Comments
01	Provider login	Provider credentials	Access provider dashboard	Dashboard loaded with booking overview	Pass	Real-time notifications active
02	View booking requests	-	Pending bookings displayed	3 pending bookings shown with customer details	Pass	Booking request time stamps accurate
03	Approve booking	Select booking, click "Approve"	Booking confirmed, player notified	Booking status changed to "Confirmed"	Pass	Automatic confirmation email sent
04	Update booking status	Mark as "Completed"	Status updated, payment processed	Status updated, payment released to provider account	Pass	Payment settlement working correctly
05	Generate booking report	Date range: Current month	PDF report with all bookings generated	Report generated: 28 bookings, Revenue: LKR 245,000	Pass	Report included charts and analytics

Module: Player Registration & Profile

Module Name	Player Registration & Profile
Test Case ID	1
Priority Level	High
Description	Verify player registration and profile setup
Prerequisites	Registration page accessible
Environment	Web Browser

Test Case Table

ID	Test Steps	Test Input	Expected Result	Actual Result	Status	Comments
01	Navigate to player registration	-	Registration form loads successfully	Form loaded with 6 sections	Pass	Mobile-responsive design working
02	Enter personal details	Name: Dasun Shanaka, Age: 28, Phone: 0717894563	Form validation passes	All fields accepted, age validation working	Pass	Phone number format validation active
03	Set sports preferences	Football	Preferences saved correctly	Preferences saved, related content suggested	Pass	Recommendation engine functioning
04	Complete skill assessment	Skill level: Intermediate, Position: Goal Keeper	Assessment results saved	Skill score: 72%, Level: Intermediate confirmed	Pass	Assessment algorithm accurate
05	Submit registration	-	Account created, welcome email sent	"Welcome to LankaArena!" email received"	Pass	Player ID: PLYR_2025_0893 generated

Module: Venue Booking Management

Module Name	Venue Booking Management
Test Case ID	2
Priority Level	High
Description	Verify venue search and booking process
Prerequisites	Player logged in, venues available
Environment	Web Browser

ID	Test Steps	Test Input	Expected Result	Actual Result	Status	Comments
01	Player login	Player credentials	Access player dashboard	Dashboard loaded with quick booking option	Pass	Recent activity section updated
02	Search venues	Location: Colombo, Sport: Cricket, Date: 2025-04-20	Relevant venues displayed	5 cricket venues in Colombo shown	Pass	Location-based sorting working
03	Select venue and time	Venue: R. Premadasa Stadium, Date: 2025-04-20, Time: 2 PM-4 PM	Availability checked	Time slot available, price: LKR 12,000 shown	Pass	Real-time availability check accurate
04	Add participants	10 team members with emails	Participant list updated	All 10 participants added, invitations sent	Pass	Bulk email functionalit y working
05	Confirm booking	-	Booking confirmed, payment page loads	Booking ID: BKNG_0420_1567 confirmed	Pass	Calendar sync with Google Calendar working

Admin Management | IT23751514 | Herath.H.M.P.P

Description

The Admin Management System of our SportHub platform serves as the central control unit for overseeing all platform operations and maintaining system integrity. The process initiates with secure administrator authentication, where authorized personnel log in using encrypted credentials, ensuring protected access to sensitive administrative functions. Upon successful verification, the system redirects admins to a comprehensive dashboard that provides a holistic view of platform activities, user interactions, and financial transactions.
For user management, administrators can register new players, coaches, and service providers by validating their information against system requirements before permanent database storage. The system enables admins to update user profiles, modify permissions, and temporarily suspend accounts when necessary, with all changes logged for audit purposes. Additionally, administrators can permanently remove inactive accounts following a multi-step confirmation process to prevent accidental data loss.
In platform oversight, admins monitor venue listings, verify service provider credentials, and ensure compliance with platform standards. They manage financial transactions, resolve payment disputes, and generate comprehensive reports on platform performance. The system incorporates real-time analytics that help admins identify trends, monitor user engagement, and make data-driven decisions to enhance platform effectiveness.
The system maintains operational excellence through integrated validation protocols, automated alert systems, and confirmation safeguards that prevent errors and preserve data integrity. By centralizing administrative functions, SportHub ensures efficient platform management, robust security measures, and complete operational control, enabling seamless sports management ecosystem operations.

Flow Chart Link –
https://drive.google.com/file/d/13q1-XOE3VKYKXvfcqQ73uBYUavqD_7N1/view?usp=drive_link

Pseudocode

```
BEGIN ADMIN_MANAGEMENT_SYSTEM

DISPLAY "SportHub Admin Login"
INPUT admin_username, admin_password

IF VALIDATE_CREDENTIALS(admin_username, admin_password) THEN
   GENERATE auth_token
   REDIRECT_TO admin_dashboard
ELSE
   DISPLAY "Invalid credentials. Please try again."
   EXIT
ENDIF

WHILE admin_logged_in DO
   DISPLAY_DASHBOARD_MENU:
      "1. User Management"
      "2. Venue Approvals"
      "3. Financial Reports"
      "4. System Analytics"
      "5. Logout"

   INPUT admin_choice

   CASE admin_choice OF
         DISPLAY_USER_MANAGEMENT_OPTIONS:
            "a. View All Users"
            "b. Verify New Users"
            "c. Suspend/Activate Users"
            "d. Update User Roles"

         INPUT user_management_choice
         EXECUTE_USER_MANAGEMENT(user_management_choice)

         FETCH pending_venues FROM database WHERE status = 'pending'
         DISPLAY pending_venues LIST

         FOR EACH venue IN pending_venues DO
            DISPLAY venue_details
            DISPLAY "Approve (A) or Reject (R)?"
            INPUT approval_decision

            IF approval_decision = 'A' THEN
               UPDATE database SET venue_status = 'approved'
               SEND_APPROVAL_EMAIL(venue_owner_email)
            ELSE
               UPDATE database SET venue_status = 'rejected'
               SEND_REJECTION_EMAIL(venue_owner_email, rejection_reason)
            ENDIF
         END FOR

         GENERATE_FINANCIAL_REPORT(start_date, end_date)
         CALCULATE total_revenue, platform_commission, provider_payouts
         DISPLAY revenue_analytics_charts
         EXPORT_REPORT_AS_PDF()

         FETCH system_metrics:
            total_users = COUNT(users)
            active_bookings = COUNT(bookings WHERE status = 'active')
```

```
            monthly_revenue = SUM(payments WHERE month = current_month)
            platform_utilization = CALCULATE_UTILIZATION_RATE()

        DISPLAY analytics_dashboard
        GENERATE_PERFORMANCE_INSIGHTS()

        INVALIDATE auth_token
        REDIRECT_TO login_page
        EXIT WHILE LOOP

    END CASE
END WHILE

END ADMIN_MANAGEMENT_SYSTEM
```

Coach Management System | IT23599222 | Wijethunga A.T.U

Description

The Player Management System of SportHub facilitates comprehensive athlete management and engagement throughout their sports journey. The process begins with player registration, where individuals create personalized profiles by providing essential information, sports preferences, and skill levels. The system authenticates player credentials and redirects them to personalized dashboards tailored to their sporting interests and experience levels. For profile management, players can maintain updated personal information, including contact details, physical attributes, medical conditions, and emergency contacts. The system allows athletes to showcase their achievements, upload performance videos, and track their progression across different sports disciplines. Players can specify their availability, preferred playing times, and location preferences to facilitate better matching with venues and opponents. In activity management, players can search and book sports facilities based on location, availability, pricing, and amenities. The system enables athletes to form teams, invite fellow players, and manage team rosters efficiently. Players can schedule practice sessions, coordinate with coaches, and track their training progress through integrated calendar and notification systems.
The system ensures seamless player experience through intelligent recommendation engines, conflict detection mechanisms, and personalized content delivery. By empowering players with self-service capabilities and comprehensive tracking tools, SportHub enhances athlete engagement, improves skill development, and fosters a vibrant sports community

Flow Chart Link-
https://drive.google.com/file/d/1ByzBNYM9EM4sS3rBR24NTx10vswaWIXB/view?usp=drive_link

Pseudocode

```
BEGIN COACH_MANAGEMENT_SYSTEM
DISPLAY "Coach Application Form"
INPUT coach_credentials:
    personal_info, coaching_certifications
    years_experience, sports_specializations
    coaching_philosophy, hourly_rate, availability

UPLOAD supporting_documents:
    certificates, id_proof, background_check

SUBMIT_FOR_VERIFICATION()

WHILE application_status = 'pending' DO
    DISPLAY "Application under review. Please wait for admin approval."
    CHECK_VERIFICATION_STATUS_EVERY_24_HOURS()
END WHILE
```

```
IF application_approved THEN
   SETUP_COACH_PROFILE()
   GAIN_ACCESS_TO_COACH_DASHBOARD()
ELSE
   DISPLAY rejection_reason
   OPTION_TO_REAPPLY_AFTER_30_DAYS()
ENDIF

WHILE coach_logged_in DO
   DISPLAY_COACH_DASHBOARD:
      "Pending Sessions: " + pending_count
      "Active Students: " + active_students_count
      "Monthly Earnings: " + monthly_income
      "Average Rating: " + average_rating

   DISPLAY_COACH_MENU:
      "1. Manage Availability"
      "2. View Session Requests"
      "3. Track Student Progress"
      "4. Update Training Programs"
      "5. Financial Reports"
      "6. Logout"

   INPUT coach_choice

   CASE coach_choice OF
      1: // Manage Availability
         DISPLAY weekly_calendar
         SET available_time_slots
         BLOCK unavailable_periods
         SET different_rates_for peak_hours, weekends
         SAVE_AVAILABILITY()

         FETCH pending_requests FROM database
         DISPLAY request_details:
            student_profile, requested_times
            special_requirements, previous_feedback

         FOR EACH request DO
            OPTION_TO: "Accept", "Decline", "Suggest Alternative Time"
            IF ACCEPT THEN
               CONFIRM_SESSION()
               SEND_CONFIRMATION_NOTIFICATION(student)
            ENDIF
         END FOR

         SELECT student FROM active_students_list
         DISPLAY student_performance_metrics
         RECORD session_notes, skill_assessments
         SET new_training_goals
         GENERATE progress_report FOR student

         CREATE new_training_module:
            module_name, objectives, duration
            exercises, difficulty_level, equipment_needed
         ASSIGN modules TO specific_students
         TRACK module_completion_status

         GENERATE income_statement:
```

```
            session_earnings, platform_commissions
            tax_deductions, net_income
        VIEW payment_history
        DOWNLOAD invoices FOR accounting

        UPDATE last_active_time
        REDIRECT_TO login_page
    END CASE
END WHILE

END COACH_MANAGEMENT_SYSTEM
```

Operations & Equipment Inventory Management | IT23628168 | Alahakoon .A.M.I.B

Description

The Coach Management System of SportHub enables sports professionals to manage their coaching practice efficiently while connecting with aspiring athletes. The process starts with coach registration, where qualified professionals submit their credentials, certifications, and coaching specialties for platform verification. Upon approval, coaches gain access to specialized dashboards designed to streamline their coaching operations.
For profile management, coaches can create comprehensive professional portfolios showcasing their qualifications, experience, coaching philosophy, and success stories. The system allows coaches to set their availability schedules, define service packages, and establish pricing structures for different coaching programs. Coaches can upload training materials, create lesson plans, and develop customized training modules for their athletes.
In session management, coaches can receive and respond to booking requests, schedule training sessions, and manage their coaching calendar efficiently. The system enables coaches to track athlete progress, record session notes, and provide personalized feedback to help players improve their skills. Coaches can monitor attendance, track payments, and manage their coaching business finances through integrated tools.
The system maintains coaching excellence through certification verification, performance monitoring, and quality assurance mechanisms. By providing coaches with business management tools and athlete connectivity platforms, SportHub enhances coaching effectiveness, expands professional opportunities, and elevates sports training standards.

Flow Chart Link-
https://drive.google.com/file/d/1Q46nkyHZlzj9v05O9BKVkIwC2Url7PCD/view?usp=drive_link

Pseudocode

```
BEGIN EQUIPMENT_INVENTORY_SYSTEM

CREATE equipment_catalog TABLE WITH columns:
    equipment_id, equipment_name, sport_type
    quantity_available, rental_price, condition_status
    last_maintenance_date, next_maintenance_due
    location, supplier_info, purchase_date

WHILE system_active DO
    DISPLAY_INVENTORY_DASHBOARD:
        "Total Equipment: " + total_items
        "Currently Rented: " + rented_count
        "Under Maintenance: " + maintenance_count
        "Low Stock Alerts: " + low_stock_alerts_count

    DISPLAY_OPERATIONS_MENU:
        "1. Add New Equipment"
        "2. Process Rental Requests"
        "3. Maintenance Scheduling"
        "4. Stock Management"
```

```
        "5. Generate Reports"
        "6. Exit"

    INPUT operation_choice

    CASE operation_choice OF
        INPUT equipment_details:
            name, category, sport_type, quantity
            purchase_cost, rental_price, supplier
            warranty_info, expected_lifespan

        GENERATE unique_equipment_id
        CREATE equipment_record IN database
        UPDATE inventory_counters
        PRINT barcode_labels FOR new_equipment

        RECEIVE rental_requests FROM booking_system
        CHECK equipment_availability(equipment_id, rental_date)

        IF available THEN
            CALCULATE rental_cost based on duration, quantity
            GENERATE rental_agreement
            UPDATE equipment_status TO 'rented'
            SCHEDULE return_date_reminders()
        ELSE
            SUGGEST alternative_equipment OR different_dates
            ADD_TO waiting_list IF requested
        ENDIF

        RUN automated_maintenance_check:
            IDENTIFY equipment_due_for_maintenance
            CHECK equipment_usage_hours SINCE last_maintenance
            INSPECT reported_issues FROM users

        SCHEDULE maintenance_tasks:
            ASSIGN_TO maintenance_staff
            SET priority_level based on equipment_criticality
            BLOCK equipment_during_maintenance_period
            UPDATE maintenance_history_record

        MONITOR stock_levels IN real_time
        IF quantity_below minimum_threshold THEN
            GENERATE purchase_requisition
            ALERT procurement_team
            CALCULATE optimal_reorder_quantity
        ENDIF

        TRACK equipment_movement:
            check_in, check_out, transfers_between_locations
            UPDATE real_time_availability_status

        GENERATE inventory_report:
            equipment_utilization_rates
            maintenance_cost_analysis
            rental_revenue_by_category
            equipment_lifespan_analytics

        EXPORT reports FOR management_review
    END CASE
END WHILE
```

END EQUIPMENT_INVENTORY_SYSTEM

Service Providers Management | IT23759398 | Karunarathna.N.T.I.N

Description

The Operations & Equipment Inventory Management System of SportHub streamlines the management of sports equipment and operational resources across the platform. The process begins with inventory setup, where administrators catalog available sports equipment, define rental parameters, and establish maintenance schedules. The system authenticates equipment handlers and provides them with specialized interfaces for inventory control.
For equipment management, the system tracks all sports gear through their lifecycle—from procurement and storage to rental and maintenance. Administrators can add new equipment to the inventory, update item specifications, and monitor equipment availability in real-time. The system enables efficient categorization of equipment by sport type, size, condition, and rental status, ensuring optimal resource utilization.
In rental operations, the system processes equipment booking requests, calculates rental fees based on duration and equipment type, and manages pickup/return schedules. The platform tracks equipment usage patterns, monitors wear and tear, and schedules preventive maintenance to ensure equipment safety and longevity. The system generates alerts for low-stock items, maintenance due dates, and equipment requiring replacement.
The system ensures operational efficiency through barcode/RFID integration, automated inventory reconciliation, and predictive maintenance scheduling. By optimizing equipment management processes, SportHub reduces operational costs, minimizes equipment downtime, and enhances the overall sports experience for all platform users.

Flow Chart Link-
https://drive.google.com/file/d/1l9gAebshVO99T5vnUNpyJWkUr_a3DQzA/view?usp=drive_link

Pseudocode

```
BEGIN SERVICE_PROVIDERS_MANAGEMENT_SYSTEM

DISPLAY "Venue Owner Registration"
INPUT venue_owner_details:
   business_name, contact_person, email, phone
   business_registration_number, tax_id

INPUT venue_information:
   venue_name, address, location_coordinates
   sports_facilities_available, capacity, amenities
   operating_hours, pricing_structure, photos

SUBMIT_FOR_VERIFICATION()

WHILE verification_pending DO
   DISPLAY "Your venue is under review by our team."
   NOTIFY_WHEN_VERIFICATION_COMPLETE()
END WHILE

WHILE provider_logged_in DO
   DISPLAY_PROVIDER_DASHBOARD:
      "Today's Bookings: " + todays_bookings
      "Monthly Revenue: " + monthly_revenue
      "Customer Ratings: " + average_rating
      "Maintenance Alerts: " + maintenance_alerts

   DISPLAY_PROVIDER_MENU:
      "1. Manage Bookings"
      "2. Update Availability"
```

```
        "3. View Financials"
        "4. Customer Feedback"
        "5. Facility Management"
        "6. Logout"

    INPUT provider_choice

    CASE provider_choice OF
        FETCH booking_requests:
            pending_approval, confirmed, completed
        DISPLAY booking_calendar VIEW
        APPROVE_OR_DECLINE booking_requests
        SEND booking_confirmations TO customers
        MANAGE booking_modifications AND cancellations

        DISPLAY availability_calendar
        BLOCK dates FOR maintenance, private_events
        SET different_pricing FOR:
            peak_hours, weekends, holidays
        UPDATE real-time_availability_status
        SYNC_WITH main_platform_calendar()

        GENERATE revenue_reports:
            daily, weekly, monthly earnings
            booking_trends, popular_time_slots
            platform_commission_calculations
        TRACK payment_status FROM customers
        MANAGE invoices AND receipts

        VIEW customer_ratings AND reviews
        ANALYZE feedback_trends
        RESPOND_TO customer_comments
        IMPLEMENT improvement_suggestions
        MONITOR overall_satisfaction_scores

        SCHEDULE routine_maintenance
        REPORT facility_issues
        MANAGE staff_schedules
        ORDER supplies AND equipment
        UPDATE facility_photos AND information

        SAVE current_session_data
        REDIRECT_TO provider_login_page
    END CASE
END WHILE

END SERVICE_PROVIDERS_MANAGEMENT_SYSTEM
```

Player Management I IT23761278 | Maduwantha.J.G.K

Description

The Service Providers Management System of SportHub empowers venue owners and sports facility operators to manage their offerings effectively while connecting with sports enthusiasts. The process initiates with provider registration, where facility owners submit their venue details, amenities, and operational information for platform verification. Upon approval, service providers access dedicated dashboards tailored to their business needs.

For venue management, providers can create comprehensive facility profiles showcasing available sports courts, equipment, changing rooms, and other amenities. The system enables providers to set dynamic pricing based on time slots, day of the week, and demand patterns. Service providers can manage availability calendars, block dates for maintenance, and offer special promotions to attract more customers.
In booking management, providers receive and confirm booking requests, coordinate with customers, and manage facility scheduling efficiently. The system enables providers to track venue utilization, monitor customer feedback, and maintain facility standards through quality control mechanisms. Providers can process payments, generate invoices, and manage their business finances through integrated accounting tools.
The system ensures service excellence through customer rating systems, performance analytics, and quality assurance protocols. By providing service providers with business management tools and customer connectivity platforms, SportHub enhances venue utilization, improves customer satisfaction, and strengthens the sports infrastructure ecosyste

Flow Chart Link-
https://drive.google.com/file/d/1ZCwpS5WcSMAv0HTFD4g9Yt57xu_h9NYk/view?usp=drive_link

Pseudocode

```
BEGIN PLAYER_MANAGEMENT_SYSTEM

DISPLAY "New Player Registration"
INPUT player_data:
   full_name, email, phone_number, password
   date_of_birth, gender, primary_sport
   skill_level, preferred_positions, location

IF VALIDATE_PLAYER_DATA(player_data) THEN
   GENERATE player_id = "PLYR_" + TIMESTAMP + RANDOM_NUMBER
   CREATE player_profile IN database
   SEND_VERIFICATION_EMAIL(player_email)
   DISPLAY "Registration successful! Please verify your email."
ELSE
   DISPLAY "Invalid data. Please check your inputs."
ENDIF

WHILE player_logged_in DO
   DISPLAY_PLAYER_DASHBOARD:
      "Welcome, " + player_name
      "Upcoming Bookings: " + upcoming_count
      "Team Invitations: " + invitation_count
      "Coach Recommendations: " + recommendation_count

   DISPLAY_MAIN_MENU:
      "1. Book Venue"
      "2. Rent Equipment"
      "3. Find Coach"
      "4. Join Team"
      "5. View Progress"
      "6. Logout"

   INPUT player_choice

   CASE player_choice OF
         DISPLAY "Enter search criteria:"
         INPUT sport_type, location, date, time_slot
         SEARCH_AVAILABLE_VENUES(sport_type, location, date, time_slot)
         DISPLAY available_venues LIST

         SELECT desired_venue
         INPUT number_of_players, duration
```

```
        CALCULATE total_cost = venue_price * duration
        PROCEED_TO_CHECKOUT()

        DISPLAY equipment_catalog FILTER_BY sport_type
        SELECT equipment_items
        INPUT rental_duration
        CALCULATE rental_cost
        ADD_TO_CART()
        PROCEED_TO_PAYMENT()

        DISPLAY coaches FILTER_BY sport_type, location, rating
        VIEW coach_profiles, certifications, availability
        SELECT coach
        SEND_COACHING_REQUEST(selected_coach, preferred_times)

        BROWSE available_teams NEAR location
        FILTER_BY sport_type, skill_level, age_group
        REQUEST_TO_JOIN_TEAM(selected_team)

        DISPLAY performance_metrics:
           matches_played, win_rate, skill_improvement
           coach_feedback, fitness_level
        SHOW progress_charts OVER time
        GENERATE improvement_recommendations()

        UPDATE last_login_time
        REDIRECT_TO homepage
    END CASE
END WHILE

END PLAYER_MANAGEMENT_SYSTEM
```